

Josh Magnus R. Ludan

Comm 459

Finding The Center of Reddit

Introduction

Reddit is one of the world's largest online communities with over 430 million monthly active users. On Reddit, users share posts on forums called “subreddits”. These subreddits are formed around particular interests or identities. These subreddits can span from the very large and general, such as /r/news, where over 24.5 million members discuss news, to very small and specific, such as /r/upenn with 14.5 thousand. Users can participate in as many of these subreddits as they can, leading to emergent network structures as users transfer information from one subreddit to another^[19].

Reddit has played several roles in US news and politics this past decade. Most recently, the subreddit r/wallstreetbets (WSB) has been in the news for their involvement in the GameStop short squeeze. WSB is a subreddit with 2.5 million members where users discuss stocks, trading, and other financial news. In January 2021, users of WSB began buying up shares of GameStop that hedge funds were shorting. This caused the stock price of GameStop to increase dramatically, inflicting losses on those originally shorting the stock^[17]. Additionally, in the 2016 US presidential elections, the subreddit r/The_Donald was a major source of support for Donald Trump's campaign and was a large source and booster for the propagation of conspiracy theories such as QAnon and Pizzagate^[18].

Aside from these notable cases, Reddit is also commonly used by many subcultures and movements as a forum to discuss ideas. The r/antiwork subreddit for example became popular during covid as workers increasingly became more critical of their working conditions^[15]. Another example is the incel subculture through subreddits like r/incel and r/MGTOW which are devoted to the discussion of misogynistic ideas^[16].

Given Reddit's size and influence on contemporary discourse and politics, it becomes important to understand its network structure. In this paper, we seek to identify the most influential subreddits within the network for several reasons. First, this understanding would allow Reddit to better understand the dynamics of harmful speech on its platform and create better moderation tools. Second, it would allow outside observers to better characterize the different movements on the platform.

With these motivations in mind, we want to answer the following research questions by constructing a network graph of Reddit with subreddits as nodes with commentator overlap determining the edges:

- 1) Which nodes in the reddit network are highly central under different definitions of centrality?
- 2) How are the top-ranked nodes under each definition different, both qualitatively and quantitatively?
- 3) How well do complex contagions spread on reddit?

Background

Cross-Community Influence

Several studies have investigated the dynamics of how different subreddits can interact with each other. Belak et al. (2012) provides grounding for the idea of using commentator relationships as a way to measure the connection between two different forums. They show this by analyzing crossposts in another forum-based website (boards.ie) and found that forums with a high degree centrality had a high impact on other forums in the sense that those high-degree forums had content which frequently spilled over (via cross-posts) to other forums.^[1]

A study by Kumary et al. (2018) studied the phenomenon of “Brigading” on Reddit where one subreddit mobilizes against another subreddit to argue with or harass the members of that community. They found that this behavior was concentrated in such a way that only 1% of communities initiated 74% of the conflict. They also measured the similarity between different subreddits by comparing their similarities based on the tf-idf vectors of posts on that subreddit and found that brigading behavior was more likely to occur when subreddits were topically similar (ex: r/conspiracy vs. r/worldnews, or r/mensrights vs. r/againstmensrights). Finally, they demonstrate the concrete impacts of mobilization by showing that the members of a brigaded subreddit tended to be less active after an attack while the attackers become more active after gaining a presence in the brigaded subreddit. This study shows the impact of harmful community mobilizations and it also shows the concept of “topical similarity” as another way to compare subreddits.^[2]

Finally, a study by Chandrasekharan et al. (2022) shows the effect of algorithmically isolating a subreddit from the rest of the network via a subreddit “quarantine”. A subreddit quarantine is a tool employed by Reddit when they wish to punish a harmful or misbehaving subreddit without outright banning it. A quarantine limits that subreddit’s influence by making it harder to access that subreddit by adding a warning before accessing that subreddit. Additionally, a quarantine prevents a subreddit from appearing on recommendation pages and from being shown in search results. The study found that the influx of new users to quarantined subreddits dropped by at least half after the quarantine.

Subreddits as Ideological hubs & Moderation

An important factor to keep in mind when thinking about the notion of cross-community influence is the impact of moderation and user preference. Given that subreddits are constructed based on shared interests and identities, content that's popular on one subreddit may not always be popular on another. Weld et al. (2021) shows that this content bias can be seen in the news links that different subreddits share. Combining data from Media Bias Fact-Check and the links that users on different subreddits share, they found that subreddits can be classified as leaning to the left or to the right based on how frequently they share links to different partisan sites. They also found that 99% of links to “extremely biased” or “low factual” content was found on only 0.5% of subreddits, which can imply that other subreddits have strong filtering mechanisms in place.^[4]

One major factor driving the content on a subreddit is user preference in the form of upvotes/downvotes on posts which can influence how widely recommended a post is. The factor of moderation is also important however since volunteer moderators exert a high degree of control over what content is and isn't allowed on a given subreddit^[5]. The rules they create can be as arbitrary as they wish, on the subreddit r/catsstandingup for example moderators use automated tools such that any user who submits a post title or comment that doesn't say “Cat.” is automatically banned and their post removed.^[6]

When we combine the effects of user preference and moderator influence, we see phenomena such as the existence of subreddits that have similar content but are segregated based on political affiliation. Zhang et al. (2020) studied the difference between r/coronavirus and r/china_flu, which is a less-moderated version of the former subreddit. They found that when these communities first appeared, their content and user base were similar with a high rate of user overlap (~25% of users from r/china_flu posted on r/coronavirus and vice versa). As time progressed however, the user overlap between the two communities shrank and posts on r/china_flu increasingly started to contain anti-chinese rhetoric.

Additional studies on Reddit's Structure

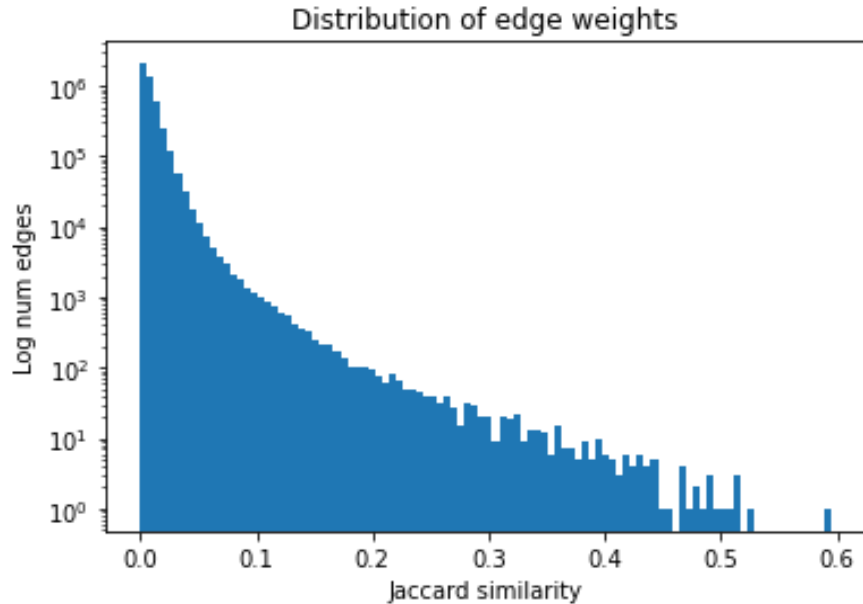
In addition to the Kumary et al. (2018) study and Zhang et al. (2020) studies above which analyzed the subreddit relationships in terms of post content and commentator overlap respectively, Chandrasekharan et al. (2018) investigated the relationships between communities based on the norms of each subreddit. In their study, they analyzed the content of moderator-deleted comments in different subreddits and found that different subreddits followed different macro and meso norms. Macro-norms are general norms that apply to most subreddits as a whole. In the 96 out of the 100 subreddits that they studied they found macro-norms against slurs, hate speech, and abusing/criticizing moderators. Meso norms are norms which only exist within a small group of subreddits. Some examples include meso-norms against ad-hominem attacks and meme/joke responses which were mostly found in debate-centered subreddits.

Olson & Neal (2015) explicitly studied the network structure of reddit by building a bipartite graph of users and subreddits where a user and a subreddit were linked together if a user commented on a subreddit and running a backbone community detection algorithm. They found that the graph exhibited a scale-free, small-world, and modular community structure. 80% of the subreddits in their dataset ($\sim 12,000$) were also too inactive to be able to maintain an edge with other subreddits in their dataset given their arbitrarily selected alpha value, meaning the graph as a whole is disconnected with a small subset of subreddits being highly connected.

Dataset and Methods

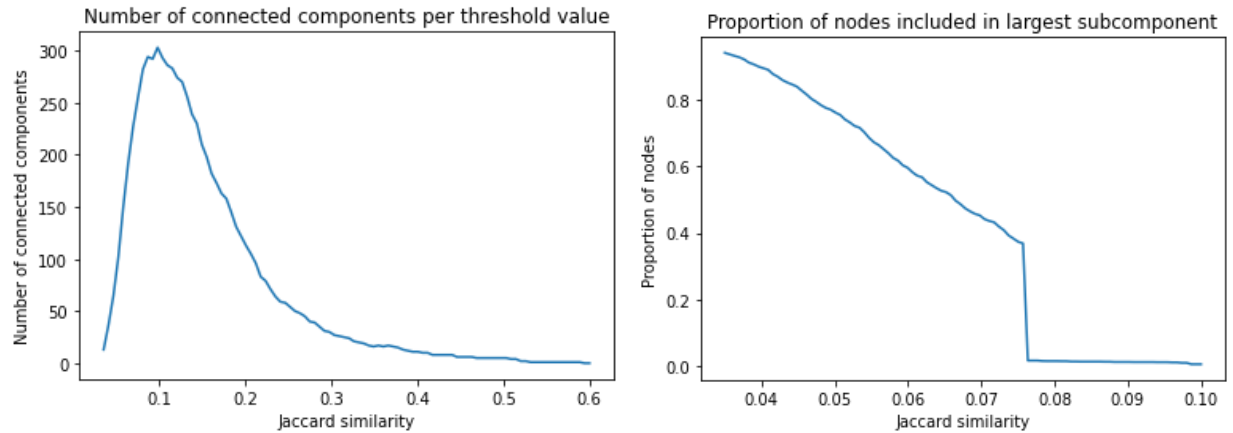
In this study, we utilize an archive of reddit from Pushshift^[10] that contains scraped comment data for comments made in June 2021. Overall, the archive contains 41,499,323 comments made by 10,482,311 users across 290,067 subreddits. Due to computational limitations, we filtered the subreddits included in the graph to remove any subreddits with less than 10,000 comments, effectively filtering for only subreddits that meet the activity threshold of having at least 350 comments a day. After filtering, we are left with only 3,023 subreddits.

To construct the links between subreddits, we represent each subreddit as a set containing the users who commented on that subreddit and take the Jaccard similarity between each subreddit's set of commentators. In the end, we obtain the following graph showing the distribution of jaccard similarities for all $3,023^2$ edges between each subreddit.



To prepare this data for use in the graph algorithms we will use, we have to select a threshold value α . This threshold value dictates which edges will be included in the graph. Edges corresponding to a jaccard similarity above α will be included while those below it will be

discarded. The selection of α has a large control over the final result, however we do not have any good priors for how to select α in our context. Given that we desire to create a good network visualization of Reddit's structure, we decided to select our value of α in such a way that it is as large as possible but also so that the largest connected subcomponent of the graph still contains 99% of all the original nodes. The two graphs below show the number of subcomponents at differing values of α as well the fraction of the nodes included in the largest subcomponent for that value of alpha. In the left graph we can see that the number of connected nodes peaks around $\alpha=0.1$ before sharply declining as more and more connected components become singleton clusters. In the right graph we can see the proportion of nodes in the largest subcomponent of the graph gradually declining as we increase the jaccard similarity before crashing around $\alpha=0.0075$.



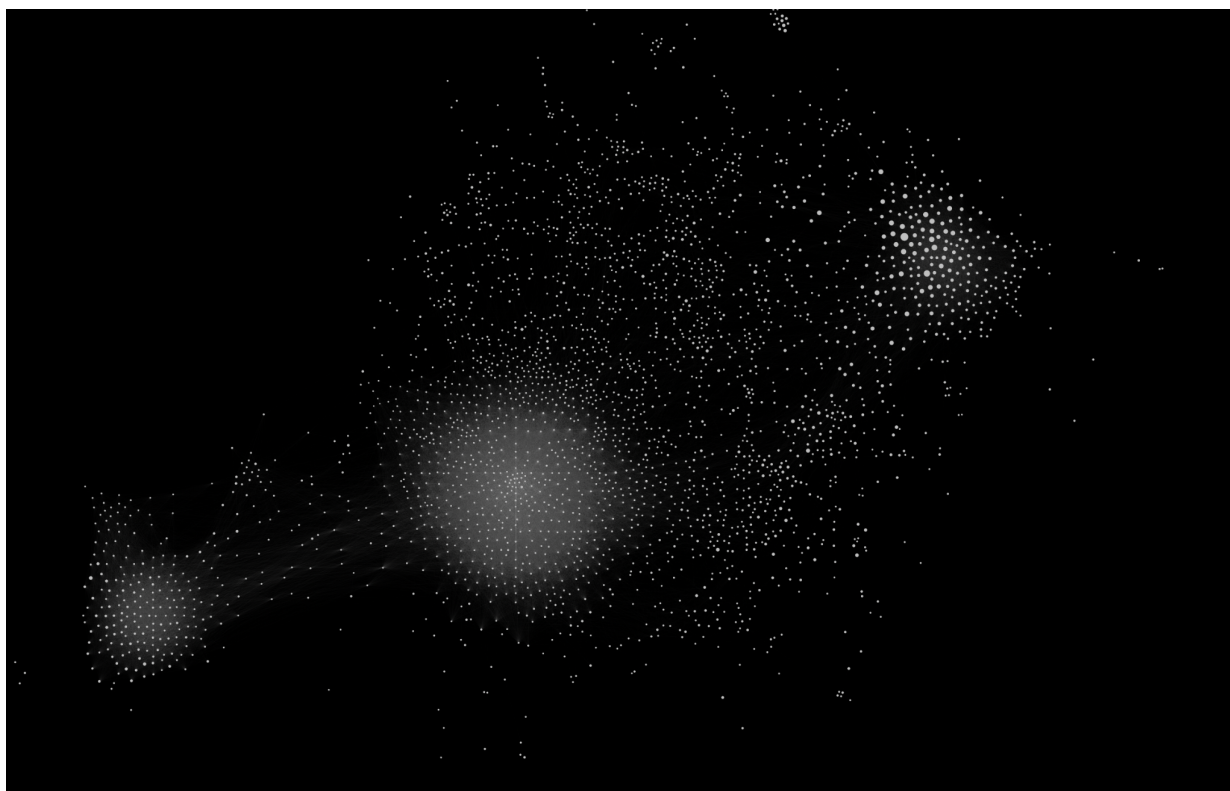
Under those desiderata, we select $\alpha = 0.035$ which yields 97,261 edges. We can interpret this to mean that we construct an edge between two subreddits if they share, at the minimum, 3.5% of their commentators.

After obtaining the graph, we then use code from Guilbeault & Centola (2021) to obtain the degree, eigenvector, betweenness, percolation, and complex centrality of the graph. In measuring the complex centrality, we initialize the threshold for each node in two ways. In the first way, we select them to be a random fraction between 0.1 and 0.5 and in the other we set the threshold to be 5^[11]. We also utilize the same code to identify the number of adopters that would arise if a complex contagion were to start using that node and its neighbors as the seed. To visualize the graph, we use Gephi^[12] utilizing the force atlas 2 layout.

Results

Overall Structure

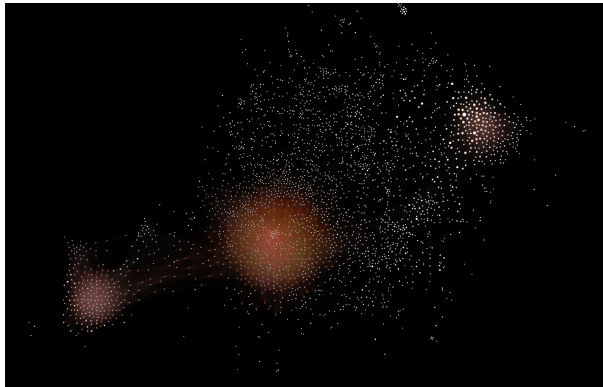
Graphing the network using Gephi yields the graph below, where nodes are sized based on the number of users within them. We note the existence of two main “clouds”. The cloud on the lower left is composed of NSFW (porn) subreddits while the larger cloud on the right are the rest of the subreddits. We note that within the cloud on the right there are two main clusters. On the upper right we can see the “default” subreddits (r/askreddit, r/gifs, etc.) which contain the most popular, mainstream subreddits. Prior to 2017, users were subscribed to most of these subreddits automatically upon account creation^[13]. The lower left cluster is more unexpected. It contains subreddits with high degree centrality and modest (~10k-100k) subscriber counts such as r/ElectricScooters, r/flatearth, and r/CVS. Compared to users in other subreddits, users in this cluster tend to have a higher number of comments in the dataset and tend to participate in more subreddits. This leads us to interpret this middle cluster as containing “niche but active” subreddits. A higher-resolution version of this graph including labels, highlighted based on user activity, can be found [here](#)



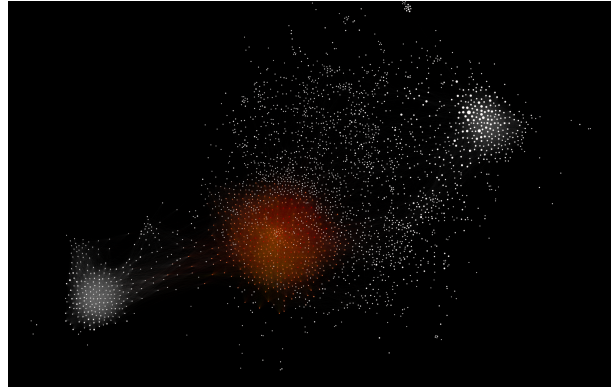
Centrality Results

These images show the graph under different highlighting rules. The more red a node is in each graph, the higher it ranks in terms of that centrality. In the top left image for example, we can see that nodes in the center tend to have the highest degree of centrality.

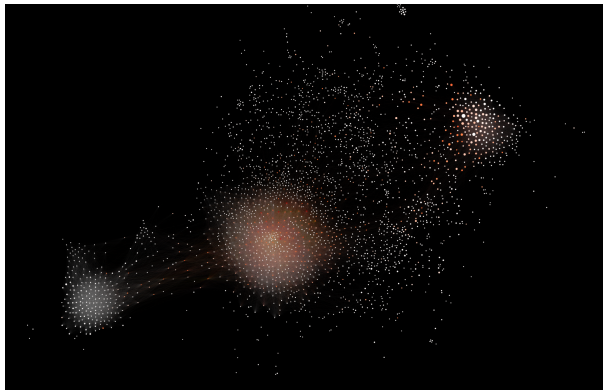
Degree Centrality



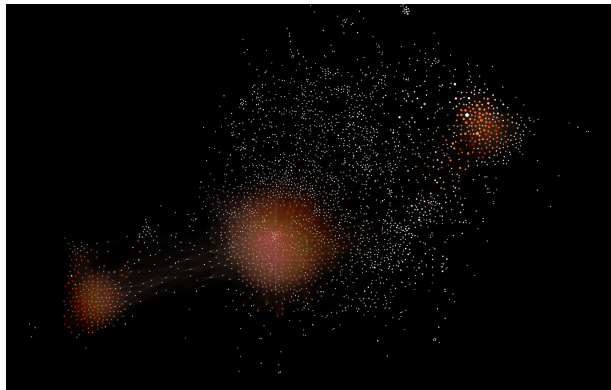
Eigenvector Centrality



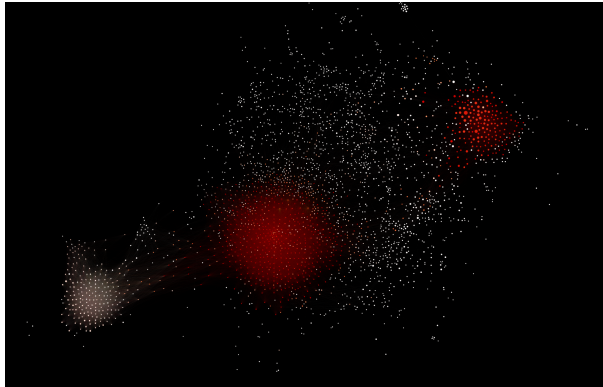
Betweenness Centrality



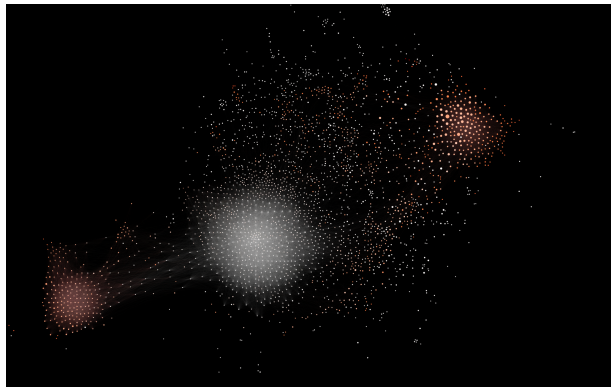
Percolation Centrality



Complex Centrality (Variable Threshold)



Complex Centrality (Threshold = 5)



The table below shows the top-ranked subreddits under each notion of centrality. The scores for each centrality are normalized such that each score is divided by the maximum centrality score in that category.

Complex (Var) Centrality		Complex (T=5) Centrality		Degree Centrality	
Mstriggahappy3	1.000	Awwducational	1.000	robloxhackers	1.000
AceAttorney	0.980	GetMotivated	0.968	memesITA	0.908
JerkBuds	0.980	ConvenientCop	0.966	Nepal	0.906
phr4r	0.972	onlyfansadvice	0.963	IndianSkincareAddicts	0.880
phinvest	0.970	AnimalsBeingBros	0.963	InstaCelebsGossip	0.873
newsokunomoral	0.969	Zoomies	0.956	akagas	0.851
AlineFariaa	0.968	woahdude	0.951	Indiangirlsontinder	0.815
lebanon	0.966	AbandonedPorn	0.950	ashhventure	0.809
Otomelsekai	0.966	DIY	0.935	NANIKPosting	0.806
AstralProjection	0.965	nevertellmetheodds	0.934	IndianStreetBets	0.804
Betweenness Centrality		Eigenvector Centrality		Percolation Centrality	
robloxhackers	1.000	robloxhackers	1.000	Nepal	1.000
Nepal	0.796	IndianSkincareAddicts	0.984	ToiletPaperUSA	0.970
memesITA	0.463	memesITA	0.982	Polska	0.913
Indiangirlsontinder	0.443	akagas	0.967	Ni_Bondha	0.888
InstaCelebsGossip	0.391	InstaCelebsGossip	0.966	BreedingMaterial	0.888
ashhventure	0.293	Indiangirlsontinder	0.961	greentext	0.881
IndianSkincareAddicts	0.289	NANIKPosting	0.959	robloxhackers	0.868
akagas	0.276	IndianStreetBets	0.957	OnlyFans101	0.845
WKHS	0.261	robloxjailbreak	0.954	boobs	0.839
WorldPacks	0.238	MurderMystery2	0.945	adorableporn	0.836

Analyzing the information from both the images and the tables, we can see that eigenvector centrality and betweenness centrality end up picking the same groups of subreddits as being highly central in the subreddit network. We can see this based on the top subreddits for each measure in the table as well as in the images where the two measures have strong highlights in the middle cluster and almost nowhere else.

For the degree and percolation centrality, we see from the image that they have highlights for the NSFW cluster, the middle cluster, and the default subreddit cluster, with a slight preference for the NSFW and center cluster over the default subreddit cluster.

We note a big difference in the highlights for complex centrality depending on how we select the node threshold values. Under a variable threshold where nodes randomly need to have 10-50% of their neighbors adopting, we see that the middle and the default subreddit clusters are highly

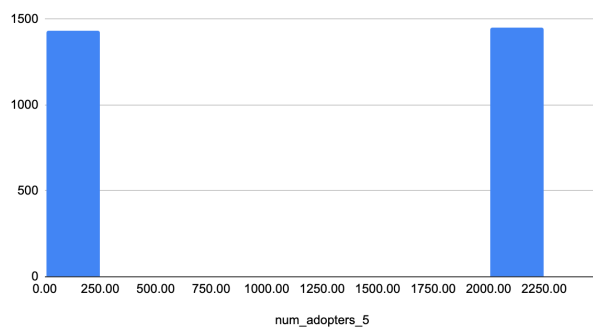
central. Under a threshold where exactly 5 neighbors need to adopt, we see that the default and the NSFW cluster is highly central.

Spread of Complex Contagions

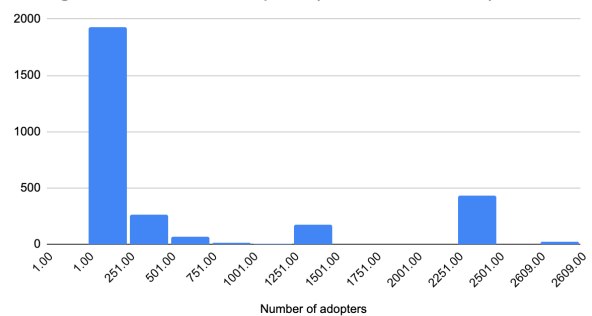
The two histograms below demonstrate how many nodes would eventually adopt a complex contagion if we seeded a complex contagion at every node. Under a complex contagion with a fixed threshold of 5 (meaning that a node will adopt a behavior if at least 5 of its neighbors adopt), we see that in half the nodes, seeding that node and its neighbor with a complex contagion would result in that complex contagion being able to reach 2022 out of 2888 other nodes in the graph. In the other half of nodes, the complex contagion would barely be able to escape its immediate neighborhood.

Under a complex contagion with a variable fraction threshold between 0.1 - 0.5 (meaning that every node would need 10% to 50% of its neighbors to adopt before itself adopting), we see multiple bins. In most of the nodes, we see that a complex contagion won't be able to spread past its neighborhood. In a small fraction of nodes (~300) we can see that seeding that node and its neighborhood would allow 100-500 other nodes to be reached. Beyond that ~250 nodes have the capability of spreading a complex contagion to 1250-1500 nodes. ~500 nodes have the capability of spreading a complex contagion to ~2250-2500 nodes. Finally, a small fraction (20) of nodes have the capability of spreading complex contagions to 2609 out of the 2888 nodes.

Histogram of number of adopters (Threshold = 5)



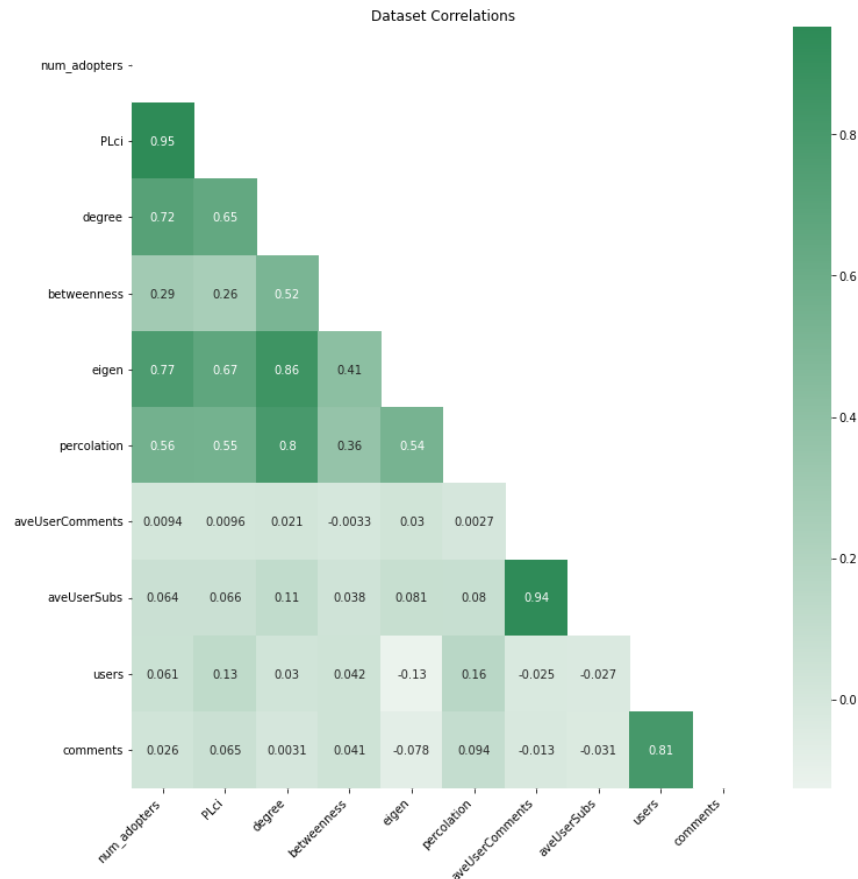
Histogram of number of adopters (Variable Threshold)



The first histogram reveals a simple fact about the connectivity of the graph: some nodes are connected enough to the main body of the graph, yet peripheral enough, that they're able to spread a complex contagion with a small, fixed threshold. The second histogram reveals the fact that different regions have different capabilities in how well they can spread a complex contagion. In fact, we can map the results of this histogram directly onto the centrality images. The region of the histogram with the low adoption rates corresponds to the nodes which don't belong in a cluster or nodes in the NSFW cluster which are relatively more disconnected from the rest of the graph. The middle region of the histogram with ~300 nodes reaching 1250-1500 nodes corresponds to the default cluster, and finally, the region in the right of the histogram where nodes reach almost all of the graph corresponds to nodes in the middle cluster.

Correlations between variables

We can also examine the correlations between the different variables. Within the network, we observe a high degree of correlations between the number of adopters and the complex centrality of a node. Corroborating with our observations above, we also note a high degree of correlation between degree and percolation centrality.



In examining the correlation between the basic statistics of a subreddit (how many users it has, how many comments it has), we see a slight correlation between the number of users a subreddit has and complex/percolation centrality. This indicates that subreddits with a high user count tend to be more central under those measures. Examining the characteristics of the users in each subreddit using variables such as such as the user activeness based on how many comments that user made overall (aveUserComments) and user participation based on how many other subreddits a user is in (aveUserSubs), we see a slight correlation between aveUserSubs and degree centrality. This indicates that if a subreddit has users that tend to be subscribed to a lot of subreddits, its degree centrality would be higher. This makes sense considering that the measure we used to form connections (jaccard similarity) would tend to favor subreddits with active user bases.

Discussion

The Usefulness of Subreddit Centrality

Using subreddits and their relation to each other as an empirical network, we were able to determine how the central nodes shifted depending on which measure of centrality we were using. Assuming that the relationships we derived were sound, the centralities we derived for the network would allow us to answer valuable questions for a variety of stakeholders interested in the structure of Reddit.

Knowing which subreddits have a high complex centrality can be valuable for researchers with limited resources in narrowing down which subreddits to focus on in their study. Studies such as the one by Farrell et al. (2019) which study the spread of misogyny on reddit through language use would be able to use the network we constructed. In their study, they selected 10 subreddits in Reddit's "manosphere" which includes subreddits like r/MGTOW, r/badwomensanatomy, and r/IncelsWithoutHate due to the fact that these subreddits had the word "incel" in their title or subreddit description. Their study characterizes the use of terms associated with misogyny such as "femoid", "cuck" or "hypergammy" in these specific communities. In addition to our results giving them more information for subreddit selection, combining the results of our research with their study leads to the natural extension of questioning how "central" these subreddits are and how effectively the misogynistic language used in these subreddit can spread to other subreddits.

Relatedly, for content moderation teams on Reddit, knowing which subreddits are more central can also be useful in identifying subreddits that might be more likely to "infect" other subreddits with problematic content. In the context of the current study, we found that the subreddit r/ElectricScooters has a high degree centrality, as well as a high complex centrality along with several of its neighbors. If somehow, harmful behaviors start from r/ElectricScooters and its immediate neighborhood, Reddit can use the map and the centralities we derived to figure out which key nodes they need to either quarantine or monitor. Historically, Reddit has been criticized for not being proactive in their moderation and only punishing subreddits once enough harm has been done^[21]. Tracking subreddits based on their position and centrality in the network graph allows Reddit to proactively track the potential for harm to spread in a manner that's independent from post content, which is what they typically use to gauge the potential for hate speech^[22].

Finally, for those interested in online communities and their impact on society, centrality measures can be useful in identifying online communities which have a large influence on Reddit. For activists seeking to influence a movement, focusing on targeting "large" and "popular" subreddits may end up being less cost effective compared to targeting smaller, but more central subreddits.

Limitations

The validity of the centralities we derived in this paper are threatened by three main potential problems.

Use of Jaccard Similarity

In constructing the edges between graphs, we used the user commentator overlap between two subreddits in terms of Jaccard Similarity. We note that this approach systematically undercounts the jaccard similarity involving larger subreddits. In the extreme case, take subreddit A with 1 million users and subreddit B with 10 thousand users, all of which also are members of subreddit A. Using Jaccard Similarity, we would find that subreddit A would have almost no overlap (0.01) with subreddit B, obscuring the fact that subreddit A would potentially have an overwhelming effect on subreddit B given that all users of subreddit B would also see the content on subreddit A. While potentially computationally expensive, this problem can be fixed by changing the denominator in the Jaccard similarity so that it represents a given target community and adapting all the algorithms used such that they use weighted, directed edges instead of the uniform, undirected edges we used in this study.

The use of Jaccard Similarity, and by extension, undirected, unweighted edges in this study has likely had a large effect on centrality. The subreddit Askreddit with 35 million members, for example, has a degree centrality almost 8 times less than the subreddit with the largest degree centrality r/RobloxHackers which only has 45k members.

Influence of Moderation

We did not account for the influence of moderation on the structure of the network. The fact that moderators can delete posts, ban users, and set the rules of their subreddit gives them a lot of power in shaping the content of their subreddit. This in turn can influence which users are likely to participate in their subreddit and how likely users from other subreddits are to engage with their subreddit. Our network has the implicit assumption that one link is as good as any other link which is false given that links into highly moderated subreddits would likely have less of an impact.

This issue is partially offset in the random-threshold complex centrality model given the fact that we can model strength of moderation by increasing that node's threshold. This however would violate the idea that we want to stick to Reddit's true network structure as possible by introducing randomness in the form of randomly set thresholds.

An accurate model that factors in moderation for setting node thresholds is likely to downrank the complex centrality of "extremist" subreddits further, due to the fact that their neighboring subreddits are likely to resist the influence coming from the extremist subreddits.

Strength of Community Influence

Our model for community influence assumes that an edge either exists or it doesn't between two nodes. Two subreddits with a Jaccard similarity of 0.5 and two subreddits with a Jaccard similarity of 0.05 are essentially treated as having the same connection strength in our model. A better model would need to employ algorithms that would be able to take tie strength into account. Given that this might potentially be computationally difficult however, our model can be improved as-is by determining a more grounded value for alpha. We essentially selected alpha in this paper out of a desire to generate a good overview of Reddit that contains its major components. It's possible to improve our estimates for centrality by having a good grounding for alpha selection. For example, it might be possible to compare the value of the jaccard similarity between subreddit vs the semantic similarity between the posts in those subreddits to determine a better "cutoff" value for alpha that corresponds to when subreddits start to share content.

Conclusion

In this paper, we sought to study the structure of Reddit by representing it as a network where subreddits are nodes and subreddit relationships are edges. To do this, we formed relationships between two subreddits based on their user overlap and then constructed the graph using Gephi. We then analyzed the centrality of the resulting graph to understand which subreddits are more central and how centrality measures can be used in the context of subreddit studies.

We found that different measures of centrality yielded different results. Under degree, eigenvector, betweenness, percolation, and complex centrality with a variable threshold, we found that a cluster of highly active but medium-sized subreddits were highly central to reddit while under degree, percolation, and complex centrality with a small fixed threshold, a set of NSFW subreddits were central to Reddit. Finally, we obtain the interesting result that only under complex centrality with a small fixed threshold was the set of most popular subreddits central to Reddit.

Our work provides a novel view of Reddit and its structure that can be used in conjunction with other studies to better understand the nature of online communities and their relationship to one another.

References

- [1] Belák, V., Lam, S., & Hayes, C. (2012). Cross-community influence in discussion fora. In *Proceedings of the International AAAI Conference on Web and Social Media* (Vol. 6, No. 1, pp. 34-41).
- [2] Kumar, S., Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2018, April). Community interaction and conflict on the web. In *Proceedings of the 2018 world wide web conference* (pp. 933-943).
- [3] Chandrasekharan, E., Jhaver, S., Bruckman, A., & Gilbert, E. (2022). Quarantined! examining the effects of a community-wide moderation intervention on reddit. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 29(4), 1-26.
- [4] Weld, G., Glenski, M., & Althoff, T. (2021). Political Bias and Factualness in News Sharing across more than 100,000 Online Communities. *ICWSM*.
- [5] <https://www.newamerica.org/oti/reports/everything-moderation-analysis-how-internet-platforms-are-using-artificial-intelligence-moderate-user-generated-content/case-study-reddit/>
- [6] <https://www.reddit.com/r/CatsStandingUp/>
- [7] Zhang, J. S., Keegan, B. C., Lv, Q., & Tan, C. (2020). A tale of two communities: Characterizing reddit response to COVID-19 through/r/China_Flu and/r/Coronavirus.
- [8] Chandrasekharan, E., Samory, M., Jhaver, S., Charvat, H., Bruckman, A., Lampe, C., ... & Gilbert, E. (2018). The Internet's hidden rules: An empirical study of Reddit norm violations at micro, meso, and macro scales. *Proceedings of the ACM on Human-Computer Interaction*, 2(CSCW), 1-25.
- [9] Olson, R. S., & Neal, Z. P. (2015). Navigating the massive world of reddit: Using backbone networks to map user interests in social media. *PeerJ Computer Science*, 1, e4.
- [10] <https://files.pushshift.io/reddit/>
- [11] Guilbeault, D., & Centola, D. (2021). Topological measures for identifying and predicting the spread of complex contagions. *Nature communications*, 12(1), 1-9.
- [12] <https://gephi.org/>
- [13] https://www.reddit.com/r/changelog/comments/65x4a7/testing_a_new_sign_up_experience/
- [14] Centola, D. (2018). How behavior spreads. In *How Behavior Spreads*. Princeton University Press.
- [15] <https://www.bloomberg.com/news/articles/2021-11-24/antiwork-subreddit-leads-boycott-on-amazon-on-black-friday-2021>
- [16] <https://journals.uic.edu/ojs/index.php/spir/article/view/12171/10299>
- [17] <https://www.bloomberg.com/news/articles/2021-01-25/gamestop-short-sellers-reload-bearish-bets-after-6-billion-loss>

- [18] https://www.dailydot.com/debug/reddit-bans-r-cbts_stream/
- [19] <https://www.redditinc.com/blog/reddits-2019-year-in-review/>
- [20] Farrell, T., Fernandez, M., Novotny, J., & Alani, H. (2019, June). Exploring misogyny across the manosphere in reddit. In Proceedings of the 10th ACM Conference on Web Science (pp. 87-96).
- [21] <https://time.com/6121915/reddit-international-hate-speech/>
- [22] https://www.reddit.com/r/announcements/comments/hi3oht/update_to_our_content_policy/